

Software Test Plan — Web Based ML Model Evaluator

Version 1.0 | Team 14 | Derived from SRS v1.0 (07-09-2026) and SADS v1.0 (14-09-2026)

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1. Purpose and Scope

Purpose. Define the verification activities required to demonstrate conformance of the Web Based ML Model Evaluator with the approved SRS and SADS, including functional, API, non-functional, and security requirements.

In scope. Authentication; CSV upload and validation; configuration; five built-in algorithms; evaluation and metrics; visualization; PDF/CSV export; per-user history; error handling; security controls; performance and concurrency targets.

Out of scope. User-supplied pretrained models, distributed training, MLOps pipelines, and offline operation.

References. SRS v1.0; SADS v1.0; OWASP ASVS; OWASP API Security Top 10.

2. Test Strategy

Level	Coverage	Tools	Owner
Unit	Metrics, validators, algorithm registry/adaptor, password hashing, token issue/verification	pytest, pytest-cov	Developers
Integration	API → validation → configuration → orchestrator → database; report generation	pytest; FastAPI/Flask test client; SQLite	Developers
System / E2E	Upload → configure → evaluate → results browser flow	Playwright or Selenium	QA
API Contract	Status codes, response schemas, JSON error envelope	Postman/Newman or pytest + requests	QA
Non-functional	Performance, concurrency, usability, portability	Locust/k6; Lighthouse; manual QA	QA
Security	Authentication/authorization, upload abuse, rate limiting, TLS, log hygiene	pytest; OWASP ZAP; manual review	QA / Security

Key strategy controls:

- Validate metric correctness against Iris and Breast Cancer Wisconsin reference datasets using a fixed random state and independently computed scikit-learn results.
- Test backend validation independently of frontend validation.
- For every API case, verify both HTTP status and documented JSON error code.

3. Test Environment and Data

Item	Configuration	
Backend	Python 3.11+; FastAPI or Flask; pandas; NumPy; scikit-learn	
Database	SQLite for development/CI; PostgreSQL for staging/deployment	
Browsers	Current Chrome, Firefox, Edge; tablet viewport 768×1024	
Network	HTTPS/TLS 1.2+ in staging; HTTP permitted locally only	
CI	Unit and integration tests on every push; E2E and security scans nightly	
ID	Dataset / Input	Purpose
D1	Iris — 150 rows, multiclass	Multiclass metrics
D2	Breast Cancer Wisconsin — binary	Binary metrics and ROC-AUC
D3	Synthetic 10,000 × 50 dataset	Performance boundary
D4	CSV with missing numeric values	Imputation vs. row exclusion
D5	CSV with categorical features/target	Categorical handling
D6	~10 MB file and 10 MB + 1 byte	Upload size boundary
D7	Non-CSV files renamed .csv	Upload security
D8	Empty/header-only/ragged/duplicate-header CSVs	Malformed CSV handling
D9	CSV containing formula-injection cells	Export sanitization
D10	Single-class target / one sample per class	Split and metric edge cases

4. Entry, Exit, and Suspension Criteria

Entry: SRS and SADS baselined; build deploys successfully; D1–D10 prepared; test accounts available.

Exit: 100% of test cases executed; ≥95% pass; no open Critical/High defects; all Must-have SRS requirements traced to passing tests; unit coverage ≥80% for evaluation, validation, and authentication; all core security cases pass.

Suspension: Smoke test fails (login + upload + one evaluation), or the test database is unavailable.

5. Functional Test Cases

5.1 Authentication

ID	Scenario	Test / Input	Expected Result
TC-001	Valid registration	Register with unique username/email and strong password	201; user created
TC-002	Password hashing	Inspect stored password; register same password for two users	bcrypt hash; no plaintext; distinct salted hashes
TC-003	Valid login	POST valid credentials	200; token issued and accepted
TC-004	Logout	Login → logout → access protected history	Logout succeeds; old token returns 401
TC-005	Invalid credentials	Wrong password and unknown user	401; identical generic response
TC-005a	Duplicate registration	Register same username/email twice	400/409; no duplicate row
TC-005b	Invalid fields	Blank/short/malformed fields	400; field-level error
TC-005c	Expired/tampered token	Use expired token and modified signature	401

5.2 Upload and Validation

ID	Scenario	Test / Input	Expected Result
TC-006	Valid CSV	Upload D2	200; dataset ID, columns, preview, warnings
TC-007	Non-CSV	Upload D7	415; file not stored
TC-007a	Size boundary	Upload D6 and D6+1 byte	At limit accepted; over limit 413
TC-008	Malformed CSV	Upload D8 variants	400/422; clear error; no stack trace
TC-009	Preview accuracy	Upload D1	Preview and column names match source
TC-009a	Missing values	Upload D4	Warnings identify affected columns/counts
TC-009b	Guest upload	Upload without auth	200 if guest upload is enabled
TC-009c	Encoding/delimiter	UTF-8 BOM, quoted commas, CRLF	Parsed correctly

5.3 Configuration

ID	Scenario	Test / Input	Expected Result
TC-010	Valid target/features	Select target and features	Normalized configuration accepted
TC-010a	Target in features	Include target among features	400
TC-010b	Unknown column	Use nonexistent feature/target	400
TC-011	Missing-value strategy	Run D4 with supported strategies	Configured strategy applied and recorded
TC-012	Invalid strategy	Use unsupported strategy	400
TC-013	Algorithm list	Inspect supported algorithms	Exactly five specified algorithms
TC-014	Multi-select algorithms	Select 1, 3, and all 5	All selected algorithms execute
TC-014a	No algorithm	Empty algorithm list	400
TC-014b	Unknown algorithm	Use unknown algorithm ID	400
TC-015	Split ratio	Test valid and invalid test_size values	0.2 accepted; invalid values rejected
TC-015a	Registry extensibility	Register test stub algorithm	Available through API without API-layer changes

5.4 Evaluation and Metrics

ID	Scenario	Test / Input	Expected Result
TC-016	End-to-end training	D2 with all five algorithms	Completed run; one result per algorithm
TC-017	Metric accuracy	D1/D2 with fixed seed	Accuracy/precision/recall/F1 match reference within 1e-6
TC-017a	Multiclass averaging	Run D1	Documented averaging method applied consistently
TC-018	Confusion matrix	Run D2	Dimensions/classes/counts match reference
TC-019	ROC-AUC applicability	Compare D2 and D1	Binary value correct; multiclass handled explicitly
TC-019a	Determinism	Run identical configuration twice	Identical seeded results
TC-019b	Categorical data	Run D5	Encoded correctly or controlled 422

ID	Scenario	Test / Input	Expected Result
TC-019c	Single-class/tiny classes	Run D10	Controlled error; no unhandled exception
TC-019d	SVM probabilities	Binary SVM run	ROC-AUC remains computable
TC-019e	Feature scaling	KNN/SVM/LogReg on unscaled data	Designed preprocessing applied; no convergence crash
TC-019f	Algorithm failure isolation	Force one algorithm to fail	Failure isolated; other results remain consistent

5.5 Visualization

ID	Scenario	Test / Input	Expected Result
TC-020	Comparison chart	Run 3+ algorithms	Chart values match result table
TC-021	Confusion matrix	Run D1/D2	Correct labels and counts
TC-022	Ranking table	Run 3+ algorithms	Documented metric ordering; deterministic ties
TC-022a	Metric applicability	Multiclass run	UI identifies non-applicable ROC-AUC

5.6 Report Export

ID	Scenario	Test / Input	Expected Result
TC-023	PDF export	GET /reports/{run_id}/pdf	Valid PDF containing dataset, config, metrics, summary
TC-024	CSV export	GET /reports/{run_id}/csv	Rows match stored metrics
TC-024a	Export consistency	Compare UI, PDF, CSV	All values consistent
TC-024b	Unknown run export	Use invalid run ID	404
TC-024c	CSV injection	Export D9	Cells neutralized

5.7 History

ID	Scenario	Test / Input	Expected Result
TC-025	Authenticated history	Evaluate while logged in	Run/results linked to user
TC-025a	Guest run	Evaluate as guest; then inspect history	Guest run absent
TC-026	List history	GET /history	Only caller's runs; newest first
TC-026a	Unauthenticated history	GET /history without token	401
TC-027	Delete own run	DELETE own run	Run and dependent records removed
TC-027a	Delete another user's run	Cross-user delete	403; owner's data unchanged
TC-027b	Repeated/unknown delete	Delete twice or unknown ID	404

6. API, Non-functional, and Security Tests

6.1 API Contract and Error Handling

ID	Test	Expected Result	Pri
TC-API-01	Documented error statuses 400/401/403/404/413/415/422/429	Status and JSON error envelope {error:{code,message,request_id}}	H
TC-API-02	request_id uniqueness and log correlation	Present, unique, and traceable in logs	M
TC-API-03	Induced internal failure	500/503; no stack trace or secrets	H
TC-API-04	POST /evaluations success semantics	Consistent 200-result or 202-polling implementation	M
TC-API-05	Guest access to own evaluation/report	Own guest-scoped run only; no cross-user access	H
TC-API-06	Malformed JSON / missing fields	400	M

6.2 Non-functional Tests

ID	Requirement	Test / Pass Criterion	Pri
TC-N01	Performance	D3 with each algorithm and all five completes within the SRS-defined time target.	H
TC-N02	Usability	Five users complete upload → results unaided; keyboard operation and WCAG AA contrast verified.	M
TC-N03	Concurrency	20 simultaneous evaluations complete without cross-user data mixing; within target latency; excess requests queue safely.	H
TC-N04	Reliability	100 evaluations; restart during run; simulate DB failure. No corruption; status remains consistent.	M
TC-N05	Portability	Chrome/Firefox/Edge; desktop/laptop/tablet; clean Docker host. Functional parity.	M
TC-N06	Modularity	Static review plus TC-015a; evaluation engine remains isolated from routing/database concerns.	L

6.3 Security Tests

ID	Security Test	Expected Result	Pri
TC-SEC-001	Password protection	No plaintext passwords in DB or logs; bcrypt policy satisfied.	H
TC-SEC-002	Input sanitization	D7/D9, SQLi and XSS payloads rejected or safely escaped; no execution/leakage.	H
TC-SEC-003	Authorization / IDOR	Identity derived from token; cross-user dataset/run access returns 403/404.	H
TC-SEC-004	Transport security	HTTP redirected/refused; TLS below 1.2 refused in staging.	H
TC-SEC-005	Rate limiting	Login/register threshold returns 429; legitimate access recovers after window.	H
TC-SEC-006	Log hygiene	Request ID/status/timing logged; no passwords, tokens, or dataset contents.	H
TC-SEC-007	Uploaded content safety	Script-like files, zip bombs, very wide CSVs and long lines rejected/limited; no execution.	H
TC-SEC-008	Temporary-file lifecycle	Expired temporary files cleaned; path guessing cannot access another user's files.	M
TC-SEC-009	Role checks, if admin UI exists	Non-admin admin-endpoint access returns 403.	M
TC-SEC-010	Token handling	Cookies Secure/HttpOnly/SameSite where applicable; JWT expiry; no sensitive claims.	M

7. Requirements Traceability

Requirement	Mapped Test Cases
MLE-F-001...005	TC-001...005c
MLE-F-006...009	TC-006...009c
MLE-F-010...012	TC-010...012
MLE-F-013...015	TC-013...015a
MLE-F-016...019	TC-016...019f
MLE-F-020...022	TC-020...022a
MLE-F-023...024	TC-023...024c
MLE-F-025...027	TC-025...027b
MLE-NF-001...006	TC-N01...N06
MLE-SEC-001...005	TC-SEC-001...005; extended security cases 006–010
SADS §4.3–4.4	TC-API-01...06

8. Defect Management

Severity	Definition	Example	Target
Critical	Data breach, crash, incorrect metrics	Cross-user history access	Before next test cycle
High	Major feature failure without workaround	Export unavailable	Before release
Medium	Feature works with workaround	Chart label incorrect	Before release if time permits
Low	Cosmetic/non-blocking issue	Alignment issue	Backlog

Record defects in GitHub Issues with: test-case ID, reproduction steps, expected result, actual result, build/commit, severity, and evidence.

9. Schedule and Responsibilities

Phase	Activity	Owner
1	Unit tests for authentication, validation, and metrics alongside implementation	Module authors
2	Integration and API contract tests after endpoints are available	Backend developers
3	E2E and usability tests after the UI flow is complete	Frontend / QA
4	Load, concurrency, and security testing in staging	QA
5	Regression, exit-criteria review, and test summary	Whole team

10. Testing Risks and Mitigation

Risk	Mitigation
Open SADS decisions: categorical handling, guest storage, FastAPI vs Flask	Resolve before finalizing affected expected results; mark dependent cases provisional until resolved.
ML nondeterminism	Use fixed random states and tolerance-based metric comparisons.
Test environment differs from production	Run load tests in a production-like container/resource profile.
Limited testing time	Execute all High-priority cases first, then Medium and Low.

Document status: Formalized test plan based on the supplied Test Plan. Detailed SRS/SADS decisions remain authoritative where this plan refers to implementation-specific values or unresolved design choices.